Avinash G

Mobile: 913-730-0694 **Avinash87.java@gmail.com**

SUMMARY:

- **8+** Years of experience in analysis, design, Implementation, development, and Maintenance of various Dynamic/Static web applications using Java/J2EE technologies.
- Experience and working knowledge of **Agile** Methodology (Scrum, extreme programming and Unified process), **Rational Unified Process** and **Test driven development** (TDD).
- Extensive experience in designing professional UI web applications using front-end technologies like **HTML/HTML5, CSS/CSS3, JavaScript, XML, DHTML, XHTML, JQuery, AJAX, JSON, Angular JS, Backbone JS, Require JS, React JS** and **Bootstrap.**
- Proficient in using **AJAX** for implementing dynamic Web Pages where the content was fetched via API calls and updated the **DOM** (**JSON** Parsing).
- Experience integrating web services with web clients written using client-side MVC frameworks such as **Angular JS, EXT JS, Ember JS** and **Backbone JS**.
- Experience with server-side frameworks like **Node JS, Express JS**.
- Experience in **DOM** (Document Object Model) to use in interacting with objects in **HTML, XHTML** and **XML documents.**
- Experience with **Node JS** along mean stack applications using **Mongo DB, Ext Js** and **Angular Js**.
- Used **HTML** Helper classes for rendering **HTML** controls, used **JQuery, Ajax** to GET & POST data.
- Experience in using **ORM** frameworks like **Hibernate, IBATIS** and **JPA**.
- Application Development experience using **Core JAVA, Multi-Threading, Exception Handling, Serialization, Externalization, Collections framework, Generics etc.**
- Experience in building web applications using **SPRING** framework features like **MVC** (Model View Controller), **AOP** (Aspect Oriented Programming), **DAO** (Data Access Object), **Spring Boot, Spring Security, Spring Batch.**
- Expertise in **spring** framework, include **Spring IOC/Core, Spring DAO support, Spring ORM, Spring AOP, Spring Security, Spring MVC** and **Spring Integration.**
- Experience in using **JERSY** along with **Spring Container** to create server side restful applications.
- Experience in Web Service Technologies like **JAXP, JAXB, SOAP, WSDL, UDDI, Restful, Apache Cxf, JMS** and **Service Oriented Architecture (SOA).**
- Experienced in using **Log4j** and good expertise in application build tools like **ANT, Gradle** and **Maven**.
- Experience in using the integrated development environments like **Eclipse, Rational** Application Developer, **IBM Web Sphere Studio Application Developer** (WSAD) and **IntelliJ** for Java/J2EE development.
- Hands on experience with an in-depth level of understanding in the strategy and practical implementation of **AWS.**
- Designed highly accessible, fault tolerant and cost effective and fault systems using multiple **EC2 instances, Auto Scaling, Elastic Load Balance** and **AMIs**.
- Proficient in **TDD** (Test Driven Development) by writing test cases using **Karma, Jasmine**.
- Experience in database development using **My SQL, PL/SQL , Oracle, DB2, No SQL** and **SQL Server** environments.
- Have knowledge on **Dynamo DB,** mostly in Cloud environment.
- Good experience on **Mongo DB** scaling across data centers and/or in-depth understanding of **Mongo DB** HA strategies, including replica sets.
- In depth knowledge of **Cassandra** and hands on experience with installing, configuring and monitoring Data Stax Enterprise cluster.
- Have worked on Backup and Recovery on Oracle, MySQL, PostgreSQL databases. I have experience in setting up automated scripts to run the backup.

- Experience in writing complex **SQL Queries, Stored Procedures, Triggers, Cursors** and **Functions**
- Worked on different Application Servers like **Web Logic, Web Sphere, JBOSS Server** and Web Server like **Tomcat**.
- Led team through examining the use of **Docker** to improve the efficiency of development and scalability of data streams and containerized micro services
- Experienced in **Rabbit MQ, JMS** and **IBM MQ.**
- Good experience on Design Patterns (J2EE) like **Singleton, Factory Method, DAO, Session Facade, Front Controller, Service Locator** and **Delegate Pattern.**
- Hands on experience in developing **Selenium IDE/ Web driver/RC/Grid scripts using Java, JavaScript, Ant/ Maven build tools.**
- Developed Applications using Rule Engines, **Drools.**
- Experienced in **UNIX Shell Scripting**.
- Experience in using **Grunt** and **Gulp** for build process.
- Experienced in unit testing using **JUnit** and **Mockito**.
- Experience in using **Bower, NPM**.
- Experience in using **JIRA** how to track, plan and analyze.
- Worked on continuous integration using **Jenkins/Hudson**
- Experience in using caching frameworks like **Redis**
- Experience in using version control tools like **GIT, CVS** and **Clear Case**
- Efficient team member with excellent work/time management skills, communications and interpersonal skills and ability to work under strict deadlines**.**

TECHNICAL SKILLS:

Programming Language	Core Java, SQL, PL/SQL
Java/J2EE Technologies	Servlets, JSP, JSTL, JDBC, JMS, JNDI, RMI, EJB, Swing, AWT, Applets, Multi-threading, Java Networking
Java Frameworks	Spring MVC, Struts 2.0, JSF
Application/Web Servers	WebLogic, IBM Web Sphere, JBoss, Tomcat
ORM Frameworks	Hibernate, JSF, IBATIS
Web Technologies	JSP, JavaScript, JQuery, AJAX, XML, XSLT, XHTML, DHTML, CSS, SASS, LESS, HTML, AngularJS, Ext JS, backbone JS.
Spring Framework	Spring Core, Spring IOC, Spring Batch, Spring Boot, Spring Security
Database Server	Oracle, DB2, SQL Server, MySQL, RDBMS, PostgressSql. No SQL Data Bases like MongoDB, Cassandra.
IDE	Eclipse, RAD, WSAD, Net Beans
Web Services	REST, SOAP, JAX-WS, JAX-RPC, JAX-RS, WSDL,Axis2, UDDI., Apache CXF
Version Control	GIT, CVS, SVN
Build Tools	ANT, Maven, Gradle
Methodologies	Agile, Scrum, Water-Fall, Rapid Application Development
Modeling Tools	Microsoft Visio, Rational Rose
Operating System	Windows, Linux, UNIX
ESB Tools	Apache Camel, Mule ESB, cucumber, redhat BPM
Testing	Junit, Jasmine, Karma, Protractor, Mockito, Selenium, Jenkins, Docker, Drools.

EDUCATION:

Degree: Bachelor of Technology, Electronics and Communication Engineering, Jawaharlal Nehru Technological University, Kakinada, India.

PROFESSIONAL EXPERIENCE:

Client: State Government of New York, Albany, New York **Nov 2015- Till Date**

Title: Sr. Java/UI developer

Description: : eClaims is the New York State Workers' Compensation Board's implementation of an electronic claim reporting standard for reports of injury filings. The WCB has adopted a national standard for claims reporting from the International Association of Industrial Accident Boards and Commissions the standard uses Electronic Data Interchange, commonly known as EDI, so that data can be transmitted electronically between the WCB and its EDI Trading Partners quickly, efficiently, and cost-effectively.

Responsibilities:

- Responsible for the analysis, documenting the requirements and architecting the application based on **J2EE** standards.
- Designed dynamic and browser compatible pages using **HTML5, DHTML, CSS3, BOOTSTRAP** and **JavaScript.**
- Used **JavaScript** libraries to create **slider** and **page-turning** animations on calendar widgets.
- Involved in developing web pages using **Angular JS** Framework and familiar with various Angular JS directives and services.
- Used **Angular JS** as frameworks to create single page applications (SPA) which can bind data to specific views and synchronize data with server using **Sass, Less, Bootstrap, Angular JS**.
- Responsible for **Web UI** development in JavaScript using **JQuery, AngularJS** and **AJAX.**
- Responsible for secure batch data flow to downstream systems using **Middleware Java technologies**.
- Developed Mean Stack from scratch including **Mongo DB** server (**Mongoose Connector**) setup and **Express JS** server development.
- Developed drag and drop document editing using **React Js**.
- Developed the **JQuery** callback functions to implement asynchronous communication using **AJAX.**
- Used the **Java Collections API** extensively in the application as security protection for **XML, SOAP**, **REST** and **JSON** to make a secure Web Deployment.
- Built backend **REST API** with **Node JS, Express JS** and also Implemented modules into **Node JS** to integrate with designs and requirement.
- GUI design and development using Java **Swing, JFC** Used for open source web testing.
- Implemented different Design patterns like **DAO, Singleton Pattern** and MVC architectural design pattern of **Spring framework**.
- Worked on the **REST Web Services** and used Spring Framework and developed the Spring Features like **Spring Core, Spring IOC, Spring AOP, Spring Data Access, Spring web test, Spring MVC, Spring DAO, Spring Boot, Spring Batch, Spring Security, Spring Integration.**
- Building **RESTful** services using **Micro-services** architecture with tools such as Spring-boot.

- Developing Intranet Web Application using J2EE architecture, using **JSP** to design the user interfaces and **Hibernate** for database connectivity.
- Development of **Hibernate** objects for data fetching for batch and front-end processing.
- Designed and developed the new features in the existing application to meet the customer's satisfaction.
- Used **Hibernate**, object relational-mapping **(ORM)** solution, technique of mapping data representation from MVC model to Oracle Relational data model with **SQL**-based scheme.
- Designed and developed the user interface screens, data base design changes and changes to user access modules.
- Responsible for implementation of java integration components for **JBPM** and **DROOL**s.
- Defined a message store architecture necessary to support **drools** rule engine input processing, scalability and recovery.
- Involved in writing **Oracle PL/SQL Stored procedures, triggers** and **views**.
- Implementing **BPM** processes in existing system using both Synchronous and Asynchronous Web Services.
- Leveraged **AWS cloud services** such as **EC2**; auto-scaling; and **VPC** (Virtual Private Cloud) to build secure, highly scalable and flexible systems.
- **AWS** is used to coordinate and manage the delivery of messages to the subscribers. Client side validations and server side validations are done according to the business needs.
- Extensively used core java concepts like **collection frame work, multi threading, OOPS and Exception Handling.**
- Managed the dependency packages by **NPM/Bower**.
- Wrote **JUNIT** test cases for web servers.
- Written test cases and done Unit testing execution with **JASMINE** and **KARMA.**
- Involved in **CICD** process using **GIT, Nexus, Jenkins job creation, Maven build, Create Docker image** and **deployment in AWS** environment.
- Converted our staging and Production environment from a handful AMI's to a single bare metal host running **Docker**.
- Involved with Sprint Planning, Sprint grooming, Allocation of User Stories using **Jira.**
- Used **Log4J** API for logging and debugging.

Environment: Spring Core, Spring IOC, Spring MVC , JavaScript, JQuery, JSON, Spring DAO , Spring Boot , Spring Batch, Spring Security, Spring Integration, Spring-Hibernate integration framework, JSP, Oracle, PL/SQL, XML, AWS (EC2, VPC), Spring web flow, Drools, Maven, JUnit, REST web service, HTML, CSS, SASS, AJAX, Jira, Angular JS, NPM/BOWER.

Client: Broadcom, Sunnyvale, California **May 2014 - Oct 2015**

Title: Sr. Java/UI developer

Description: Broadcom Limited is a diversified global semiconductor leader built on 50 years of innovation, collaboration and engineering excellence. With roots based in the rich technical heritage of technology powerhouses AT&T/Bell Labs, Lucent and Hewlett-Packard/Agilent, Broadcom focuses on technologies that connect our world.

Responsibilities:

- As a Senior J2EE Developer assisted my team in developing two different Web-based applications
- Designed and developed the application using **AGILE-SCRUM** methodology.
- Used **HTML, JavaScript**, **Angular JS**, **JQuery**, **AJAX**, **CSS**, and **Bootstrap** as data and presentation.
- Developing & Customizing UI **JavaScript** Plug-in using **JQuery** and **JSON**.
- Created Mock-up designs and developed various functionalities with **JavaScript** and **JQuery**.
- Developed Model pop-ups with **Angular JS** for responsive web design on mobile devices.
- Wrote **Rest** services to feed data to an application with **AngularJS** framework.
- Developed a core component of Analytics with **Node JS**, **Backbone JS.**
- Developed user interface using **JSP** with **JavaBeans, JSTL and Custom Tag Libraries, Javascript, CSS, JQuery, Node JS, HTML, SASS and Ajax t**o speed the application.
- Used Web Services using **SOAP**, **WSDL**, **UDDI** and **JAX-WS** programming model.
- Involved in implementation of enterprise integration with Web Services and Legacy Systems using **SOAP** and **REST** (Using Axis and Jersey frameworks).
- Designed and developed **Micro-Services** using (MEAN Stack) node.js, Mongo DB.
- Developed web services using **SOAP**, **WSDL** and **Apache Axis**, which helped communicating through different modules of the application.
- Involved in creating **Restful** web services using **JAX-RS** and **JERSY** tool.
- Involved in **J2EE** Design Patterns such as **Data Transfer Object (DTO), DAO**, Value Object and Template.
- Primarily focused on the spring components such as **Spring MVC**, **Dispatcher Servlets**, **Controllers**, Model and View Objects, View Resolver.
- Used **Spring IOC**, **AOP, Spring Batch** modules to integrate with the application.
- Generated **DAO's** to map with database tables using **Hibernate.**
- Developed a multi-user web application using **JSP**, **Servlet**, **JDBC**, **Spring** and **Hibernate** framework to provide the needed functionality.
- Analyze, debug and fix issues related to **thread locking, clustering, network connections, Swing** client performance on customer's production environment
- Designed Interactive GUIs using **Java Applets and Swing/JCF**
- Developed **Spring Beans** and configured **Spring** using applicationContext.xml.
- Developing the web application using Core Java concepts like **Collections**, **OO concepts, Exception handling** for developing application modules
- Assisted Database Developers using **PL/SQL** for managing data associated with the tables in **Oracle**.
- Managed and maintained **No SQL** (**Cassandra**) databases in production domain.
- Uses Second Level Cache to enhance performance and Proper Memory Management.
- Developed application using **JMS** for sending and receiving Point-to-Point **JMS** Queue messages.
- Used **Apache CXF** as the Web Services framework to implement the **REST** APIs involved.
- Setup replication clusters with multiple nodes for **MySQL** and **PostgresSQL**.
- Database development required creation of new tables, **SQL** stored procedures, functions, views, indexes and constraints, triggers and required **SQL** tuning to reduce the response time in the application.
- Involved in designing different **BPM** patterns in order to address the complex business scenarios are part of solution and technical design.
- Built application to track unique users using **Redis** accross various data groupings.

- Created task handlers to expose the **BPM** processes to **REST** service calls in the process designer itself and integrated them with spring.
- Used **Maven** for generating system builds and **Jenkins** for continuous integration.
- Whenever a new github branch gets started, **Jenkins**, our Continuous Integration server, automatically attempts to build a new **Docker** container from it, The **Docker** container leverages Linux containers.
- Created **WAR** files from **Maven** and deployed them on **Tomcat**.
- Worked on **Drools** Rules Engine and Expectation Rule Engine for writing Business Rule Validation
- Wrote unit-testing codes using **JUnit**, resolved bugs and other defects using **Firebug**.
- Wrote **JUNIT** Test cases for **Spring Controllers** and Web Service Clients in Service Layer using **Junit**.
- Used **Log 4j** for debugging, testing and maintaining the system state.
- Using **JIRA** to manage the issues/project work flow.

Environment: Spring, Hibernate, Web services, JAX-RS, JUNIT, SOAP, UI, SQL, JSP, JDBC, JSON, JavaScript, JQuery, AJAX, Html, CSS, Angular JS, PL/SQL, PostgressSql Maven, log4j, Jenkins, JIRA, JMS, APACHE AXIS, Apache CXF, GIT, UNIX, AGILE-SCRUM, Oracle, Tomcat, Drools.

Client: NJM Insurance Group, Trenton, NJ. **Nov 2013 - Apr 2014**

Title: Sr. JAVA/J2EE Developer

Description: NJM was originally implemented in a mainframe and this project is responsible for converting from Mainframe to J2EE. The technologies used are Process Server for managing the business process and J2EE technologies for implementing the Application itself.

Responsibilities:

- Actively involved in defining and designing the tiers of the applications and implementation using **J2EE** best design practices.
- Followed **Agile** Methodology in analyze, define, and document the application, which will support functional and business requirements. Coordinate these efforts with Functional Architects.
- Developed **UI** pages with **JSP, Spring MVC, HTML5, CSS3, JavaScript, JQuery,** and used **Boot Strap** for creating Responsive Web pages.
- Implemented scripting with **AJAX** and **XML** to dynamically refresh the page contents.
- Used **JQuery** to navigate to different pages of the application using **DOM** elements, and to create **AJAX** pages.
- Used **AJAX and JSON** to process asynchronous actions between front and back end.
- Actively involving in creating **JavaScript** function for user input validation and verification.
- Coded **JavaScript** for page functionality and Pop up Screens.
- Used Web Services using **SOAP**, **WSDL**, **UDDI** and **JAX-WS** programming model.
- Well versed with core Java concepts like **Collections, Swing, Serialization and Servlets & JSP**.
- Developed **Micro-Service** to provide RESTful API utilizing **Spring Boot** with Spring MVC.

- Implemented **Spring MVC** Architecture, **Servlet** targeting request module and flow of control generating **JSP** in Presentation layer.
- Used **Spring IOC, AOP** modules to integrate with the application.
- Developed and maintained elaborate service based architecture utilizing open source technologies like **Hibernate ORM** and **Spring MVC** Framework.
- Handled rule engine with **JBoss Drools** for developing business rules of the applications.
- Used various core java concepts such as **Multi-Threading**, **Exception Handling**, **Collection API**'s to implement various features and enhancements.
- Developed **DAO** (Data Access Object) classes in order to access data from database.
- Developed **ORM** scripts for **Hibernate** to synchronize Domain Model and Relational Model.
- Developed complex Web Services and tailored **JAX-RS** API to suit the requirement.
- Involved in integration of claimant information with **IBM MQ** Resource Adapter for message queuing.
- Involved in writing the Java utility classes to convert **MQ** based control claimant information
- Installed and configured **JBOSS** server with **My SQL** persistence. Extracted rules data from the database and decision tables.
- Employed **Redis** to host centralized job queue for parallel servers to pull and distribute jobs among cluster of workers
- Implemented **RDMS** and **MySQL** as resource tier and developed **MYSQL** Stored procedures, Functions.
- Wrote unit-testing codes using **JUnit**.
- Writing Test cases using **Mockito** frameworks.
- Used **Maven** for generating system builds and **Jenkins** for continuous integration.

Environment: J2EE, Hibernate, Spring MVC Framework, UML, Servlet, JSP, JSF, AJAX, DHTML, Java Script, JQuery, JSON, EJB (Session and Message Driven Beans), Eclipse , IBM MQ, My SQL, JBoss, Drools, JUnit, Mockito, Maven, Web services, DAO, Web Sphere.

Client: Uhlig LLC, Overland Park, KS. **Nov 2012- Oct 2013**

Title: UI Developer

Description: Uhlig LLC and its subsidiaries have played a leading role in the growth of variable-content publishing, consistently redefining the traditional boundaries between print and electronic media. Today, its products range from rich, full-color periodicals to complex legal and financial documents, each of which is custom-produced for each of its readers using the company's powerful online editing and content-management tools.

Responsibilities:

- Interacting with the system analysts, business users for design & requirement clarifications.
- Development of UI models using **HTML**, **JSP**, **JavaScript**, **AJAX** and **CSS**.

- Implemented the presentation layer using **JSP**, **JSP** Tag Libraries (**JSTL**), **HTML**, **CSS** and **JQuery**. Also, Performed validations using validation framework for data inputs.
- Used **AJAX** to handle the user requests asynchronously and to have faster responses from the server by directly connecting to the server.
- Used **RESTful** services for publishing Web Service interfaces.
- Involved in writing **XML** validation classes using Struts validation framework.
- Worked on the **JAVA Collections API** for handling the data objects between the business layers and the front end.
- Developed the application using **Struts** to develop presentation tier and business layer.
- Used **Hibernate** extensively to have Database access mechanism with complex queries through **Hibernate Query**, Criteria interfaces.
- Extensively used **SQL** in constructing views, indexes, stored procedures, triggers, cursors, functions, relational database models.
- Developed Applications using Rule Engines, **Drools.** Explored **Drools** modules such as **Drools** Expert, and **Drools** Flow, which adds event processing capabilities to the platform.
- Used **Singleton**, **Service Locator**, **Data Value Objects**, and **Data Access Objects** Design Patterns in interacting between presentation, service and database components.
- Used **WebSphere** application server for deploying the application
- Used **IBM MQ** Series for messaging services for the users who want to become as registered users.
- Wrote various types of Business rules in the .drl i.e., drools file using **Drools** rule engine for business decision making.
- Involved in writing **JUnit** Test Cases and **ANT** Build Script.
- **SVN** version control system has been used to check-in and check-out the developed artifacts.

Environment: Java, J2EE Design Patterns, XML, CSS3, Web Sphere, Struts, Web Services, JSP, JSTL, EJB, Servlet, AJAX, Hibernate, RAD , IBM MQ Series, Rest, SQL Server, Drools, Unix, UML.

Client: ICICI Bank, Hyderabad, India **Apr 2010 - Oct 2012**

Title: Java Developer

Description: ICICI Bank is an Indian multinational banking and financial services company. It offers a wide range of banking products and financial services for corporate and retail customers through a variety of delivery channels and specialized subsidiaries in the areas of investment banking, life, non-life insurance, venture capital and asset management. Some other information like account payee details are maintained in the online banking database**.**

Responsibilities:

- Developed the front end browser based screens with JAVA and J2EE technologies like **JSP, Servlets**, **JavaScript**, **HTML** and **CSS** form presentation layer.
- Developed web application using **JSP**, **Servlet**, **Struts** and integrating the application using Java Messaging Service **API** and Enterprise Java Bean to integrate with the backend systems using **XML**.

- Processing **XML** files using **SAX** parsers and transformations using **XSLT**.
- Developed **JSP's** with **Struts** custom tags and implemented **JavaScript** validation of data
- Used **JDBC** to connect to the **oracle** database and **JNDI** to lookup administered objects.
- Developed **Struts** forms and actions for validation of user request data and application functionality.
- Wrote the **MDB** (Message Driven Bean) stuff to perform capture the response messages from the backend.
- Used business delegate to call **EJB** from **struts** action class.
- Designed and developed numerous Session Beans and deployed on **WebLogic** Application web Server.
- Implemented **DAO** layer using **IBATIS** and wrote queries for persisting demand Core Banking related information from the backend system using Query tool (**Oracle**).
- Developed the Login screen so that only authorized and authenticated administrators can only access the application.
- Developed Authentication and Authorization modules where authorized persons can only access the inventory related operations.
- Done automaton for build process using **ANT** Build scripts and handled all the releases and delivery to the client.

Environment: EJB, Servlet, JavaScript, Struts, Java, Web logic, XML, IBATIS, JSP, JBOSS, ANT, Eclipse, Oracle.

Client: Infotech, India. **August 2008 – Mar 2010**

Title: Java Developer

Description: Infotech Enterprises focus includes aerospace, defense, rail transportation, off-highway & industrial, power generation, mining, oil & gas, communications, utilities, geospatial, semiconductor and medical technology. We align closely with the business needs, goals, culture, and core values of our clients. This reflects in the deep, long-standing relationships we have developed and sustained with some of the leading names in these industries.

Responsibilities:

- Analyzed business requirements, participated in technical design, development and testing of various modules assigned.
- Worked on Design, UI, **HTML, CSS, JQuery**, accessibility, and optimization.
- Translated designs and style guides into functional user interfaces, ensuring cross browser compatibility, performance and robustness.
- Converted wireframes and mockups for new web interface for a network analysis virtual appliance.
- Worked with full UI control of **HTML/CSS**, maintaining code and cleanliness.
- Created some prototype to implement some of the latest features of **HTML** and **CSS**.
- Developed presentation tier as **HTML**, **JSP**s using **Struts** Framework.
- Extensively worked with making calls to APIs to fetch the data from the backend as **JSON** response and parse it on the UI for **DOM/CSS** Manipulation.

- Developed **JSPs, Servlets**, and **EJBs** in the **Struts Framework** to effectively handle the workflow of the application
- Developed the middle tier using **EJBs Stateless Session Bean, Java Servlets**.
- Entity Beans used for accessing data from the **MySQL database**.
- Implemented and Updating the database Tables, Packages, procedures, functions views
- Worked on **Hibernate** for data persistence.
- Improved page load performance by analyzing content and making adjustments to the code (by following optimization standards).
- Worked with complete redesigns of a daily events website and focused on improving readability, utility, and overall **UX.**
- Worked closely with QA team to understand the reported defects and fix them.

Environment: Java, Hibernate, Struts, EJBs, JSP, Servlet, Junit, ANT, MySQL, log4j, JSON, HTML, JavaScript, JQuery.